

Smart Bug Tracking & Test Management System

Second Project Review

LIMA K (KITM24MCA-2012)

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- **Problem:** Manual bug tracking using spreadsheets leads to inefficiency, lack of transparency, and poor collaboration between testers and developers.
- **Solution:** Centralized web-based system with role-based access (Admin, Tester, Developer).
- **Key Benefits:**
 - Real-time tracking
 - Improved communication
 - Reduction in manual errors
 - Scalable architecture

Module-Wise Description

- **Accounts Module**

- User registration with email and password.
- Role selection during registration: Admin, Tester, Developer.
- Secure login/logout functionality.
- Profile management – update personal details and password.
- Password reset via email (optional).

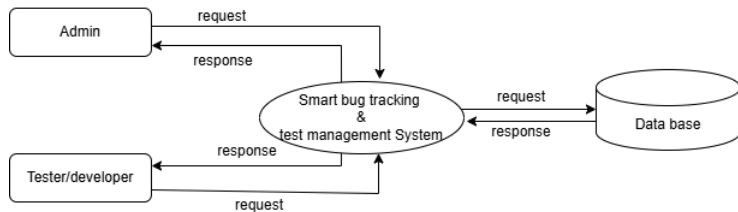
- **Dashboard Module**

- Role-specific dashboards tailored to user permissions.
- **Admin Dashboard:** Overview of system activity, user management, project statistics.
- **Tester Dashboard:** Quick access to test cases, recent bug reports, test execution status.
- **Developer Dashboard:** List of assigned bugs, pending fixes, project summaries.
- Widgets for real-time updates (e.g., new bug alerts).

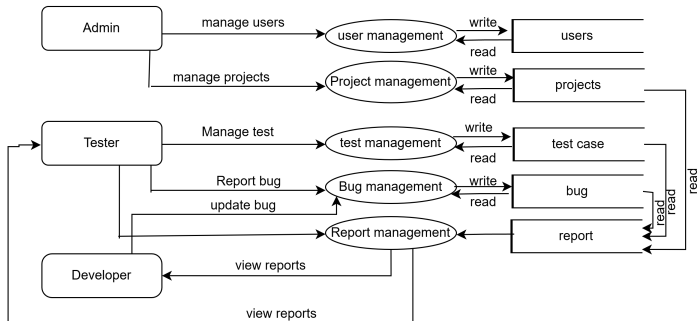
- **Test Management Module**

- Create, edit, and delete test cases with fields: title, steps, expected result, actual result, status.
- Organise test cases under projects.
- Execute test cases and record results (Pass / Fail / Blocked).
- Link test cases to reported bugs for traceability.
- View test execution history.

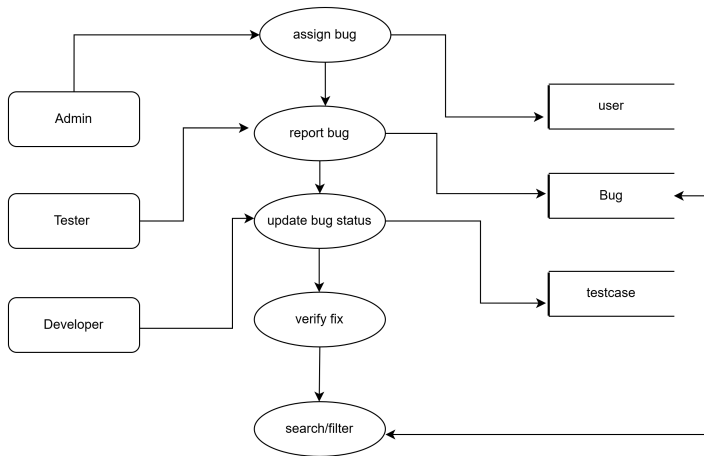
Data Flow Diagram – Level 0



DFD – Level 1



DFD – Level 2



Database Schema – Key Tables

- User Profile Table

Field Name	Data Type	Description
id	Integer (PK)	Unique identifier for user profile
user_id	Integer (FK)	References Django auth_user table
role	Varchar(20)	Role of user (Admin / Tester / Developer)

- project Table

Field Name	Data Type	Description
id	Integer (PK)	Unique project identifier
name	Varchar(100)	Name of the software project
description	Text	Brief description of the project
created_by	Integer (FK)	Admin who created the project
created_at	DateTime	Project creation date

- Testcase Table

Field Name	Data Type	Description
id	Integer (PK)	Unique test case ID
project_id	Integer (FK)	Project to which test case belongs
title	Varchar(150)	Test case title
description	Text	Test case description
steps	Text	Steps to execute the test
expected_result	Text	Expected output
priority	Varchar(10)	LOW / MEDIUM / HIGH
test_type	Varchar(15)	MANUAL / AUTOMATED
created_by	Integer (FK)	Tester who created the test case
created_at	DateTime	Test case creation date

- Test execution Table

Field Name	Data Type	Description
id	Integer (PK)	Unique execution ID
test_case_id	Integer (FK)	Test case being executed
executed_by	Integer (FK)	Tester who executed the test
execution_type	Varchar(15)	MANUAL / AUTOMATED
status	Varchar(10)	PASS / FAIL
actual_result	Text	Actual outcome after execution
executed_at	DateTime	Execution timestamp

- Bug Table

Field Name	Data Type	Description
id	Integer (PK)	Unique bug ID
project_id	Integer (FK)	Related project
test_case_id	Integer (FK, Nullable)	Related test case
title	Varchar(150)	Bug title
description	Text	Detailed bug description
severity	Varchar(15)	Low / medium / high / critical
status	Varchar(20)	Open / in_progress / fixed / closed
reported_by	Integer (FK)	Tester who reported the bug
assigned_to	Integer (FK)	Developer assigned to fix
created_at	DateTime	Bug reported date

- Report Table

Field Name	Data Type	Description
id	Integer (PK)	Unique report ID
project_id	Integer (FK → Project.id)	Project reported
created_by	Integer (FK → User.id)	User who generated report
report_date	DateTime	Date of report generation
summary	Text	Brief report summary
details	Text	Optional detailed report

Product Backlog

ID	Priority	Size (hours)	Sprint	Status	Name
1	Medium	8	1	Planned	Registration
2	Medium	8	—	Planned	Login
3	High	30	2	Partially completed	Table Design
4	High	58	3		Coding
5	Medium	5	4	Planned	Testing Data
6	High	5	5	Planned	Output generation

Table: Product Backlog (Sprints 1–3 are active; sprints 4–5 are future)

Table 1: User Stories for Smart Bug Tracking System

USER STORY ID	AS A (TYPE OF USER)	I WANT TO	SO THAT I CAN
1	Tester	create a new test case with steps and expected results	document test scenarios clearly
2	Tester	report a bug and attach screenshots	developers can understand the issue quickly
3	Developer	view all bugs assigned to me	prioritize my work effectively
4	Developer	update a bug's status to "Resolved" with a fix description	testers can verify the fix
5	Admin	manage user accounts and roles	only authorized personnel have access to the system

Sprint 1

Backlog item	Status and completion date	Original estimate
Django setup	06/01/2026	2
User registration/login	06/01/2026	3
Role redirection	08/01/2026	2
Basic templates	08/01/2026	3
Total		10

Sprint 2

Backlog item	Status and completion date	Original estimate
CRUD for test cases	14/01/2026	4
Link test cases to projects	15/01/2026	3
Bug report form	06/02/2026	3
Bug listing	06/02/2026	4
Total		14

Sprint 3 (In Progress)

Backlog item	Status and completion date	Original estimate
Assign bugs	19/02/2026	2
Developer view	27/02/2026	3
Status updates	27/02/2026	4
Commenting	27/02/2026	2
Total		11

Completed Modules:

- **Accounts Module** Registration and login implemented with role-based access (Admin, Tester, Developer).
- **Dashboard Module** Role-specific dashboards developed with basic navigation and data display.
- **Test Management Module** Test case creation, listing, and execution implemented. Linking test cases to bugs enabled.
- **Bug Tracking Module** Bug reporting and listing completed. Assignment and status update workflow in progress.
- **Reporting Module** Initial design planned. Implementation to be completed in the next phase.

Working Features:

- Registration / login with role selection
- Tester can create/view test cases and report bugs
- Developer can see assigned bugs (static assignment for now)
- Admin can view all users

Screenshots:

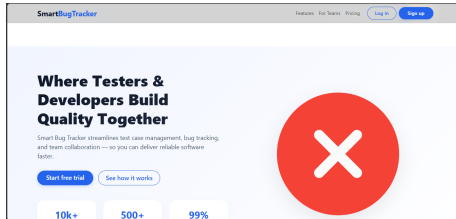


Figure: home page

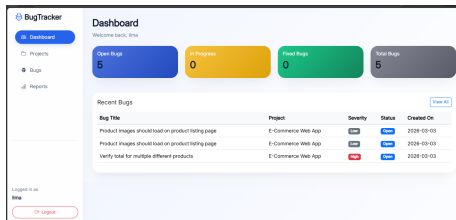


Figure: tester dashboard

- Real-time notifications (email / in-app) for bug assignments and status changes.
- Advanced search & filters for bugs and test cases.
- Graphical reports (bug trends, test execution statistics).
- Integration with version control (Git) to link commits to bug fixes.
- Mobile-responsive UI.
- Automated test case execution (future scope).

- **Current Status:** Core modules functional; bug workflow under active development.
- **Key Achievements:**
 - Role-based access implemented
 - Test case and bug creation working
 - Agile methodology followed with clear sprint goals
- **Next Steps:**
 - Complete bug assignment and status transitions
 - Add test execution logging
 - Begin reporting module

Thank You!

Questions?